

The Kalman filter may have a false confidence in the certainty of its state estimate. However, for the fading-memory filter, $K_k > 0$ for all k , and the filter will always be responsive to new measurements. A larger value of α will make the filter more responsive to new measurements. In the limit as $\alpha \rightarrow \infty$, we see from Equation (7.123) that $K_\infty = 1$. This will result in a measurement update from Equation (7.124) of

$$\begin{aligned}\hat{x}_k^+ &= \hat{x}_k^- + (1)(y_k - \hat{x}_k^-) \\ &= y_k\end{aligned}\tag{7.125}$$

In other words, the fading-memory filter, when carried to an extreme, ignores the system model and estimates the state solely on the basis of the measurements. This is the same thing that will happen if the process noise is extremely large. The Kalman filter will ignore the system model because we are telling it that we do not have any confidence in the system model.

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7.5 CONSTRAINED KALMAN FILTERING

In the application of state estimators, there is often known information that does not fit into the Kalman filter equations in an obvious way. For example, suppose that we know (on the basis of physical considerations) that the states satisfy some equality constraint $Dx = d$, or some inequality constraint $Dx \leq d$, where D is a known matrix and d is a known vector. This section discusses some ways of incorporating those constraints into the Kalman filter equations.

Some researchers have treated state equality constraints by reducing the system model parameterization [Wen92], and this will be discussed in Section 7.5.1. Others have handled state equality constraints by treating them as perfect measurements [Por88, Hay98], and this will be discussed in Section 7.5.2. A third approach is to incorporate the state constraints into the derivation of the Kalman filter [Chi85, Sim02], and this will be presented in Section 7.5.3. A final approach is to incorporate the constraints by discarding that portion of the pdf of the state estimate that violates the constraints [Shi98, Sim06b], and this will be discussed in Section 7.5.4.

7.5.1 Model reduction

Some researchers have treated state equality constraints by reducing the system model parameterization [Wen92]. This is straightforward but there are some disadvantages with this approach. First, it may be desirable to maintain the form and structure of the state equations due to the physical meaning associated with each state. The reduction of the state equations makes their interpretation less natural and more difficult. Second, equality constraints that are formulated this way cannot be extended to inequality constraints. On the other hand, the model reduction approach is conceptually straightforward and usually can be easily implemented.

As an example of the model reduction approach, consider the system

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & -2 & 2 \end{bmatrix} x_k + \begin{bmatrix} w_{k1} \\ w_{k2} \\ w_{k3} \end{bmatrix} \\ y_k &= \begin{bmatrix} 2 & 4 & 5 \end{bmatrix} x_k + v_k \end{aligned} \quad (7.126)$$

Now suppose that we also know, on the basis of our understanding of the physics underlying the problem, that the following constraint is always satisfied between the states:

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix} x_k = 0 \quad (7.127)$$

This means that $x_k(3) = -x_k(1)$. If we make this substitution for $x_k(3)$ in the original state and measurement equations, we obtain

$$\begin{aligned} x_{k+1}(1) &= x_k(1) + 2x_k(2) - 3x_k(1) \\ &= -2x_k(1) + 2x_k(2) \\ x_{k+1}(2) &= 3x_k(1) + 2x_k(2) - x_k(1) \\ &= 2x_k(1) + 2x_k(2) \\ y_k &= 2x_k(1) + 4x_k(2) - 5x_k(1) + v_k \\ &= -3x_k(1) + 4x_k(2) + v_k \end{aligned} \quad (7.128)$$

These equations can be written in matrix form as

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} -2 & 2 \\ 2 & 2 \end{bmatrix} x_k + \begin{bmatrix} w_{k1} \\ w_{k2} \end{bmatrix} \\ y_k &= \begin{bmatrix} -3 & 4 \end{bmatrix} x_k + v_k \end{aligned} \quad (7.129)$$

We have reduced the filtering problem with equality constraints to an equivalent but unconstrained filtering problem. An advantage of this approach is that the dimension of the problem has been reduced, and so the computational effort of the problem is less. One disadvantage of this approach is that the physical meaning of the state variables has been lost. Also, this approach can only be used for equality constraints (i.e., constraints of the form $Dx = d$) and cannot be used for inequality constraints (i.e., constraints of the form $Dx \leq d$).

7.5.2 Perfect measurements

Some researchers treat state constraints as perfect measurements (i.e., no measurement noise) [Por88, Hay98]. Suppose that our constraints are given as $Dx_k = d$, where D is a known $s \times n$ matrix ($s < n$), and d is a known vector. We can solve the constrained Kalman filtering problem by augmenting the measurement equation with s perfect measurements of the state:

$$\begin{aligned} x_{k+1} &= F_k x_k + w_k \\ \begin{bmatrix} y_k \\ d \end{bmatrix} &= \begin{bmatrix} H_k \\ D \end{bmatrix} x_k + \begin{bmatrix} v_k \\ 0 \end{bmatrix} \end{aligned} \quad (7.130)$$

The state equation is the same as usual, but the measurement equation has been augmented. The fact that the last s elements of the measurement equation are

noise free means that the Kalman filter estimate of the state will always be consistent with these s measurements; that is, the Kalman filter estimate will always be consistent with the constraint $D\hat{x}_k^+ = d$. Note that the new measurement noise covariance will be singular – the last s rows and the last s columns of the measurement noise covariance will be zero. A singular covariance matrix does not present any theoretical problems [Gee97]. In fact, Kalman’s original paper [Kal60] presents an example that uses perfect measurements. However, in practice a singular covariance increases the possibility of numerical problems [May79, p. 249], [Ste94, p. 365]. In addition, the use of perfect measurements is directly applicable only to equality constraints. It can be extended to inequality constraints by adding small nonzero measurement noise to the “perfect” measurements, but then the constraints will be soft [Mah04a] and it will be difficult to control how close the state estimate gets to the constraint boundary.

7.5.3 Projection approaches

Another approach to constrained filtering is to incorporate the state constraints into the derivation of the Kalman filter [Chi85, Sim02]. We can incorporate the constraints into a maximum probability derivation of the filter, or a mean square derivation of the Kalman filter. Also, we can simply project the standard Kalman filter estimate onto the constraint surface.

7.5.3.1 Maximum probability approach Assuming that x_0 , w_k , and v_k are Gaussian, the Kalman filter solves the problem

$$\hat{x}_k = \operatorname{argmax}_{x_k} \operatorname{pdf}(x_k | Y_k) \quad (7.131)$$

That is, $\hat{x}_k$ is the value of x_k that maximizes $\operatorname{pdf}(x_k | Y_k)$. In the above equation, Y_k is the vector of measurements up to and including time k ; that is, $Y_k = [y_1^T \cdots y_k^T]^T$. This interpretation of the Kalman filter looks at x_k as a random variable with a pdf that is conditioned on the measurements up to and including time k . The Kalman filter estimate is that value of x_k that maximizes its conditional pdf. If the noise processes are Gaussian, then

$$\operatorname{pdf}(x_k | Y_k) = \frac{\exp[-(x_k - \bar{x}_k)^T P_k^{-1} (x_k - \bar{x}_k)/2]}{(2\pi)^{n/2} |P_k|^{1/2}} \quad (7.132)$$

where n is the dimension of the state, P_k is the covariance of the state estimate, and $\bar{x}_k$ is defined as the mean of x_k conditioned on the measurements Y_k :

$$\bar{x}_k = E(x_k | Y_k) \quad (7.133)$$

To maximize $\operatorname{pdf}(x_k | Y_k)$, we can maximize $\ln \operatorname{pdf}(x_k | Y_k)$, which means minimizing $(x_k - \bar{x}_k)^T P_k^{-1} (x_k - \bar{x}_k)$. Now suppose that we have the additional constraint that $Dx_k = d$. The solution of this constrained minimization problem is the constrained state estimate $\tilde{x}$. That is,

$$\tilde{x}_k = \operatorname{argmin}_{\tilde{x}_k} (\tilde{x}_k - \bar{x}_k)^T P_k^{-1} (\tilde{x}_k - \bar{x}_k) \text{ such that } D\tilde{x}_k = d \quad (7.134)$$

Constrained optimization problems can be solved using the Lagrange multiplier method discussed in Section 11.2 [Ste94, Moo00]. We form the Lagrangian L and

find the necessary conditions for a minimum as follows:

$$\begin{aligned} L &= (\tilde{x}_k - \bar{x}_k)^T P_k^{-1} (\tilde{x}_k - \bar{x}_k) + 2\lambda^T (D\tilde{x}_k - d) \\ \frac{\partial L}{\partial \tilde{x}} &= P_k^{-1} (\tilde{x}_k - \bar{x}_k) + D^T \lambda = 0 \\ \frac{\partial L}{\partial \lambda} &= D\tilde{x}_k - d = 0 \end{aligned} \quad (7.135)$$

where λ is the n -element Lagrange multiplier. Solving these equations gives

$$\begin{aligned} \lambda &= (DP_k D^T)^{-1} (D\bar{x}_k - d) \\ &= (DP_k D^T)^{-1} (D\hat{x}_k - d) \\ \tilde{x}_k &= \bar{x}_k - P_k D^T \lambda \\ &= \hat{x}_k - P_k D^T (DP_k D^T)^{-1} (D\hat{x}_k - d) \end{aligned} \quad (7.136)$$

We see that the constrained state estimate $\tilde{x}$ is equal to the unconstrained state estimate $\hat{x}$, minus a correction term.

7.5.3.2 Least squares approach Another way to solve the constrained Kalman filtering problem is to approach the problem from a least squares point of view. In this approach, we find the constrained state estimate $\tilde{x}$ as

$$\tilde{x} = \operatorname{argmin}_{\tilde{x}} E(\|x - \tilde{x}\|^2 | Y) \text{ such that } D\tilde{x} = d \quad (7.137)$$

where we have dropped the time subscripts for ease of notation. This interpretation of the Kalman filter looks at x as a random variable. The quantity $(x - \tilde{x})$ (for any constant $\tilde{x}$) is also a random variable. The conditional expected value can be written as

$$\begin{aligned} E(\|x - \tilde{x}\|^2 | Y) &= \int (x - \tilde{x})^T (x - \tilde{x}) \operatorname{pdf}(x | Y) dx \\ &= \int x^T x \operatorname{pdf}(x | Y) dx - 2\tilde{x} \int x \operatorname{pdf}(x | Y) dx + \tilde{x}^T \tilde{x} \end{aligned} \quad (7.138)$$

We form the Lagrangian for the constrained optimization problem as

$$\begin{aligned} L &= E(\|x - \tilde{x}\|^2 | Y) + 2\lambda^T (D\tilde{x} - d) \\ &= \int x^T x \operatorname{pdf}(x | Y) dx - 2\tilde{x} \int x \operatorname{pdf}(x | Y) dx + \tilde{x}^T \tilde{x} + \\ &\quad 2\lambda^T (D\tilde{x} - d) \end{aligned} \quad (7.139)$$

Assuming that x_0 , w_k , and v_k are Gaussian, the standard Kalman filter estimate $\hat{x}$ is given by

$$\begin{aligned} \hat{x} &= E(x | Y) \\ &= \int x \operatorname{pdf}(x | Y) dx \end{aligned} \quad (7.140)$$

Solving the constrained minimization problem involves setting the partial derivatives of the Lagrangian of Equation (7.139) equal to zero. This gives the equations

$$\begin{aligned} \frac{\partial L}{\partial \tilde{x}} &= -2\hat{x} + 2\tilde{x} + 2D^T \lambda = 0 \\ \frac{\partial L}{\partial \lambda} &= D\tilde{x} - d = 0 \end{aligned} \quad (7.141)$$

Solving these equations for λ and $\tilde{x}$ gives

$$\begin{aligned}\lambda &= (DD^T)^{-1}(D\hat{x} - d) \\ \tilde{x} &= \hat{x} - D^T(DD^T)^{-1}(D\hat{x} - d)\end{aligned}\quad (7.142)$$

We see that the constrained state estimate $\tilde{x}$ is equal to the unconstrained state estimate $\hat{x}$, minus a correction term. This is similar to the constrained estimate that was obtained by the maximum probability approach in Equation (7.136).

7.5.3.3 General projection approach A third way to derive the constrained state estimate is to begin with the standard unconstrained estimate $\hat{x}$ and project it onto the constraint surface $Dx = d$. This can be written as

$$\tilde{x} = \operatorname{argmin}_{\tilde{x}} (\tilde{x} - \hat{x})^T W (\tilde{x} - \hat{x}) \text{ such that } D\tilde{x} = d \quad (7.143)$$

where W is any positive definite weighting matrix. [W is chosen to weight various elements of the difference $(\tilde{x} - \hat{x})$. This is generally based on the designer's relative confidence in the elements of the unconstrained state estimate.] The solution to the above problem is

$$\tilde{x} = \hat{x} - W^{-1}D^T(DW^{-1}D^T)^{-1}(D\hat{x} - d) \quad (7.144)$$

This is the most general approach to the problem. Note that the maximum probability estimate of Equation (7.136) is equal to this if we set $W = P^{-1}$. The mean square estimate of Equation (7.142) is equal to this if we set $W = I$.

It is shown in [Chi85, Sim02] that the constrained state estimate of Equation (7.144) has several interesting properties.

1. The constrained estimate is unbiased. That is, $E(\tilde{x}) = E(x)$.
2. Setting $W = P^{-1}$ results in the minimum variance filter. That is, if $W = P^{-1}$ then $\operatorname{Cov}(x - \tilde{x}) \leq \operatorname{Cov}(x - \hat{x})$ for all $\hat{x}$.
3. Setting $W = I$ results in a constrained estimate that is always (i.e., at each time step) closer to the true state than the unconstrained estimate. That is, if $W = I$ then $\|x_k - \tilde{x}_k\|_2 \leq \|x_k - \hat{x}_k\|_2$ for all k .

The projection approach to constrained filtering has the advantage that it can be easily extended to inequality constraints. That is, if we have the constraints $Dx \leq d$, then the constrained estimate can be obtained by modifying Equation (7.143) and solving the problem

$$\tilde{x} = \operatorname{argmin}_{\tilde{x}} (\tilde{x} - \hat{x})^T W (\tilde{x} - \hat{x}) \text{ such that } D\tilde{x} \leq d \quad (7.145)$$

The problem defined above is known as a quadratic programming problem [Fle81, Gil81]. There are several algorithms for solving quadratic programming problems, most of which fall in the category known as active set methods. An active set method uses the fact that it is only those constraints that are active at the solution of the problem that are significant in the optimality conditions. Assume that we have s inequality constraints (i.e., D has s rows), and q of the s inequality constraints are active at the solution of Equation (7.145). Denote by $\hat{D}$ and $\hat{d}$ the q rows of D and q elements of d corresponding to the active constraints. If the correct

set of active constraints was known *a priori* then the solution of Equation (7.145) would also be a solution of the equality constrained problem

$$\tilde{x} = \operatorname{argmin}_{\tilde{x}} (\tilde{x} - \hat{x})^T W (\tilde{x} - \hat{x}) \text{ such that } \hat{D}\tilde{x} = \hat{d} \quad (7.146)$$

This shows that the inequality constrained problem of Equation (7.145) is equivalent to the equality constrained problem of Equation (7.146). Therefore, all of the properties of the equality constrained state estimate enumerated above also apply to the inequality constrained state estimate. Standard quadratic programming routines² can be used to solve inequality constrained problems that are in the form of Equation (7.145).

■ EXAMPLE 7.11

Suppose that we have an unconstrained estimate and covariance given as

$$\begin{aligned} \hat{x} &= \begin{bmatrix} 3 \\ 3 \end{bmatrix} \\ P &= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned} \quad (7.147)$$

That is, we are twice as certain of our x_2 estimate as we are of our x_1 estimate. We also know (from our understanding of the underlying system) that the state must satisfy the constraint

$$\begin{aligned} Dx &= d \\ \begin{bmatrix} 1 & 1 \end{bmatrix} x &= 1 \end{aligned} \quad (7.148)$$

Clearly, the unconstrained estimate does not satisfy this constraint. The least squares approach to constrained estimation uses Equation (7.142) to compute the constrained estimate as

$$\begin{aligned} \tilde{x}_{LS} &= \hat{x} - D^T(DD^T)^{-1}(D\hat{x} - d) \\ &= \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)^{-1} \left(\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} - 1 \right) \\ &= \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} \end{aligned} \quad (7.149)$$

We see that the estimates for x_1 and x_2 both changed by the same amount (from the unconstrained values of 3, to the constrained values of 1/2). The maximum probability approach to constrained estimation uses Equation (7.136) to compute the constrained estimate as

$$\begin{aligned} \tilde{x}_{MP} &= \hat{x} - PD^T(DPD^T)^{-1}(D\hat{x} - d) \\ &= \begin{bmatrix} -1/3 \\ 4/3 \end{bmatrix} \end{aligned} \quad (7.150)$$

The estimate for x_1 changed by 10/3 (from the unconstrained value of 3, to the constrained value of -1/3). The estimate for x_2 changed by 5/3. We see

²For example, the QP function in MATLAB's Optimization Toolbox.

that the estimate for x_1 changed twice as much as the estimate for x_2 , because the certainty of the unconstrained x_2 estimate was twice the certainty of the unconstrained x_1 estimate. This is illustrated in Figure 7.1.

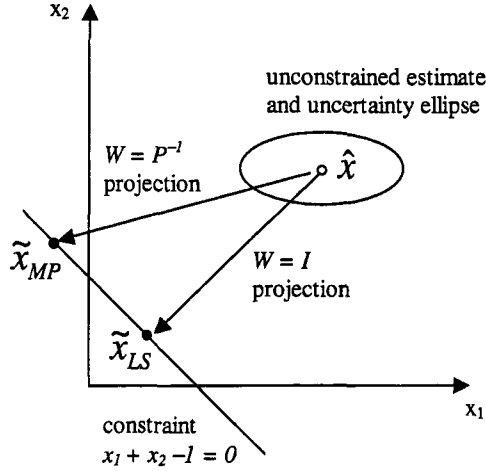


Figure 7.1 In Example 7.11, the unconstrained estimate violates the equality constraint. The least squares approach to constrained estimation projects the estimate in the direction perpendicular to the constraint surface. The maximum probability approach projects the estimate in the direction P^{-1} relative to the constraint surface.

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7.5.4 A pdf truncation approach

In the projection approach to constrained estimation discussed in the previous section, the state estimates are projected onto the constraint surface. In the pdf truncation approach, we take the probability density function that is computed by the Kalman filter (assuming that it is Gaussian) and truncate it at the constraint edges. The constrained state estimate then becomes equal to the mean of the truncated pdf [Shi98, Sim06b]. This approach is designed for inequality constraints on the state, although it can also be applied to equality constraints.

Suppose that at time k we have the s scalar state constraints

$$a_{ki} \leq \phi_{ki}^T x_k \leq b_{ki} \quad i = 1, \dots, s \quad (7.151)$$

where $a_{ki} < b_{ki}$. This is a two-sided constraint on the linear function of the state $\phi_{ki}^T x_k$. If we have a one-sided constraint, then we set $a_{ki} = -\infty$ or $b_{ki} = +\infty$. Now suppose at time k that we have a standard Kalman filter estimate $\hat{x}_k$ with covariance P_k . The problem is to truncate the Gaussian pdf $N(\hat{x}_k, P_k)$ at the s constraints given in Equation (7.151), and then find the mean $\tilde{x}_k$ and covariance $\tilde{P}_k$ of the truncated pdf. These new quantities, $\tilde{x}_k$ and $\tilde{P}_k$, become the constrained state estimate and its covariance.

In order to make the problem tractable, we will define $\tilde{x}_{ki}$ as the state estimate after the first i constraints of (7.151) have been enforced, and $\tilde{P}_{ki}$ as the covariance

of $\tilde{x}_{k_i}$. We therefore initialize

$$\begin{aligned} i &= 0 \\ \tilde{x}_{k_i} &= \hat{x}_k \\ \tilde{P}_{k_i} &= P_k \end{aligned} \quad (7.152)$$

Now perform the following transformation:

$$z_{k_i} = \rho W^{-1/2} T^T (x_k - \tilde{x}_{k_i}) \quad (7.153)$$

ρ is an orthogonal $n \times n$ matrix that will be determined later, and T and W are obtained from the Jordan canonical decomposition of $\tilde{P}_{k_i}$. This transformation will allow us to solve the pdf truncation problem that we have posed, and find the mean of the pdf as the estimated state after i constraints have been enforced. From the description of T and W we know that

$$TWT^T = \tilde{P}_{k_i} \quad (7.154)$$

T is orthogonal and W is diagonal (therefore, its square root is very easy to compute). Next we use Gram-Schmidt orthogonalization [Moo00] to find the orthogonal ρ matrix that satisfies

$$\rho W^{1/2} T^T \phi_{k_i} = [(\phi_{k_i}^T \tilde{P}_{k_i} \phi_{k_i})^{1/2} \quad 0 \quad \dots \quad 0]^T \quad (7.155)$$

The Gram-Schmidt orthogonalization procedure for computing ρ is given as follows.

1. Suppose that ρ is an $n \times n$ matrix with rows ρ_i ($i = 1, \dots, n$):

$$\rho = \begin{bmatrix} \rho_1 \\ \vdots \\ \rho_n \end{bmatrix} \quad (7.156)$$

The first row of ρ is computed as

$$\rho_1 = \frac{\phi_{k_i}^T T W^{1/2}}{(\phi_{k_i}^T \tilde{P}_{k_i} \phi_{k_i})^{1/2}} \quad (7.157)$$

2. For $k = 2, \dots, n$, perform the following.

- (a) Compute the k th row of ρ as follows:

$$\rho_k = e_k - \sum_{i=1}^{k-1} (e_k^T \rho_i) \rho_i \quad (7.158)$$

where e_k is the unit vector; that is, e_k is an n -element column vector comprised entirely of zeros, except that its k th element is a 1.

$$e_k = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow k\text{th element} \quad (7.159)$$

(b) If the ρ_k computed above is zero, then replace it with the following:

$$\rho_k = e_1 - \sum_{i=1}^{k-1} (e_1^T \rho_i) \rho_i \quad (7.160)$$

(c) Normalize ρ_k :

$$\rho_k = \frac{\rho_k}{\|\rho_k\|_2} \quad (7.161)$$

It can be shown from Equations (7.153)–(7.155) that z_{k_i} has a mean of 0 and covariance matrix of identity. With these definitions we see that the upper bound of Equation (7.151) is transformed as follows:

$$\begin{aligned} \phi_{k_i}^T x_k &\leq b_{k_i} \\ \phi_{k_i}^T T W^{1/2} \rho^T z_{k_i} + \phi_{k_i}^T \tilde{x}_{k_i} &\leq b_{k_i} \\ \frac{(\phi_{k_i}^T T W^{1/2} \rho^T) z_{k_i}}{(\phi_{k_i}^T \tilde{P}_{k_i} \phi_{k_i})^{1/2}} &\leq \frac{b_{k_i} - \phi_{k_i}^T \tilde{x}_{k_i}}{(\phi_{k_i}^T \tilde{P}_{k_i} \phi_{k_i})^{1/2}} \\ [1 \ 0 \ \dots \ 0] z_{k_i} &\leq \frac{b_{k_i} - \phi_{k_i}^T \tilde{x}_{k_i}}{(\phi_{k_i}^T \tilde{P}_{k_i} \phi_{k_i})^{1/2}} \\ &\leq d_{k_i} \end{aligned} \quad (7.162)$$

where d_{k_i} is defined by the above equation. Similarly we can see that

$$\begin{aligned} [1 \ 0 \ \dots \ 0] z_{k_i} &\geq \frac{a_{k_i} - \phi_{k_i}^T \tilde{x}_{k_i}}{(\phi_{k_i}^T \tilde{P}_{k_i} \phi_{k_i})^{1/2}} \\ &\geq c_{k_i} \end{aligned} \quad (7.163)$$

where c_{k_i} is defined by the above equation. We therefore have the normalized scalar constraint

$$c_{k_i} \leq [1 \ 0 \ \dots \ 0] z_{k_i} \leq d_{k_i} \quad (7.164)$$

Since z_{k_i} has a covariance of identity, its elements are statistically independent of each other. Only the first element of z_{k_i} is constrained, so the pdf truncation reduces to a one dimensional pdf truncation. The first element of z_{k_i} is distributed as $N(0, 1)$ (before constraint enforcement), but the constraint says that z_{k_i} must lie between c_{k_i} and d_{k_i} . We therefore remove that part of the Gaussian pdf that is outside of the constraints and compute the area of the remaining portion of the pdf as

$$\int_{c_{k_i}}^{d_{k_i}} \frac{1}{\sqrt{2\pi}} \exp(-\zeta^2/2) d\zeta = \frac{1}{2} \left[\operatorname{erf}(d_i(k)/\sqrt{2}) - \operatorname{erf}(c_{k_i}/\sqrt{2}) \right] \quad (7.165)$$

where $\operatorname{erf}(\cdot)$ is the error function, defined as

$$\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t \exp(-\gamma^2) d\gamma \quad (7.166)$$

(Note that the error function is sometimes defined without the $2/\sqrt{\pi}$ factor, which can lead to confusion. However, the above definition is the most commonly used

one.) We normalize the truncated pdf so that it has an area of one, and we find that the truncated pdf (i.e., the constrained pdf of the first element of z_{k1}) is given by

$$\begin{aligned} \text{pdf}(\zeta) &= \begin{cases} \alpha \exp(-\zeta^2/2) & \zeta \in [c_{k1}, d_{k1}] \\ 0 & \text{otherwise} \end{cases} \\ \alpha &= \frac{\sqrt{2}}{\sqrt{\pi} [\text{erf}(d_{k1}/\sqrt{2}) - \text{erf}(c_{k1}/\sqrt{2})]} \end{aligned} \quad (7.167)$$

We define $z_{k,i+1}$ as the random variable that has the same pdf as z_{k1} except that the pdf is truncated and normalized, so that its pdf lies entirely between the limits c_{k1} and d_{k1} :

$$\text{pdf}(z_{k,i+1}) = \text{truncated pdf}(z_{k1}) \quad (7.168)$$

We can compute the mean and variance of $z_{k,i+1}$ as follows:

$$\begin{aligned} \mu &= E[z_{k,i+1}] \\ &= \alpha \int_{c_{k1}}^{d_{k1}} \zeta \exp(-\zeta^2/2) d\zeta \\ &= \alpha [\exp(-c_{k1}^2/2) - \exp(-d_{k1}^2/2)] \\ \sigma^2 &= E[(z_{k,i+1} - \mu)^2] \\ &= \alpha \int_{c_{k1}}^{d_{k1}} (\zeta - \mu)^2 \exp(-\zeta^2/2) d\zeta \\ &= \alpha [\exp(-c_{k1}^2/2)(c_{k1} - 2\mu) - \exp(-d_{k1}^2/2)(d_{k1} - 2\mu)] + \mu^2 + 1 \end{aligned} \quad (7.169)$$

The mean and variance of the transformed state estimate, after enforcement of the first constraint, are therefore given as

$$\begin{aligned} \tilde{z}_{k,i+1} &= [\mu \ 0 \ \cdots \ 0]^T \\ \text{Cov}(\tilde{z}_{k,i+1}) &= \text{diag}(\sigma^2, 1, \dots, 1) \end{aligned} \quad (7.170)$$

We then take the inverse of the transformation of Equation (7.153) to find the mean and variance of the state estimate after enforcement of the first constraint.

$$\begin{aligned} \tilde{x}_{k,i+1} &= TW^{1/2} \rho^T \tilde{z}_{k,i+1} + \tilde{x}_{k1} \\ \tilde{P}_{k,i+1} &= TW^{1/2} \rho^T \text{Cov}(\tilde{z}_{k,i+1}) \rho W^{1/2} T^T \end{aligned} \quad (7.171)$$

We then increment i by one and repeat the process of Equations (7.153)–(7.171) to obtain the state estimate after enforcement of the next constraint. Note that $\tilde{x}_{k0}$ is the unconstrained state estimate at time k , $\tilde{x}_{k1}$ is the state estimate at time k after the enforcement of the first constraint, $\tilde{x}_{k2}$ is the state estimate at time k after the enforcement of the first two constraints, and so on. After going through this process s times (once for each constraint), we have the final constrained state estimate and covariance at time k :

$$\begin{aligned} \tilde{x}_k &= \tilde{x}_{ks} \\ \tilde{P}_k &= \tilde{P}_{ks} \end{aligned} \quad (7.172)$$

Figure 7.2 shows an example of a one-dimensional state estimate before and after truncation. Before truncation, the state estimate is outside of the state constraints. After truncation, the state estimate is set equal to the mean of the truncated pdf. Figure 7.3 shows another example. In this case, the unconstrained state estimate is inside the state constraints. However, truncation changes the pdf and so the constrained state estimate changes to the mean of the truncated pdf.

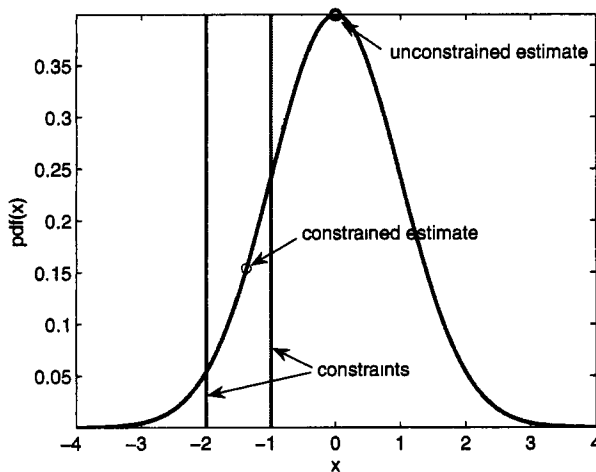


Figure 7.2 The unconstrained estimate at $x = 0$ violates the constraints. The constrained estimate, which is at $x \approx -1.38$, is the centroid of the truncated pdf.

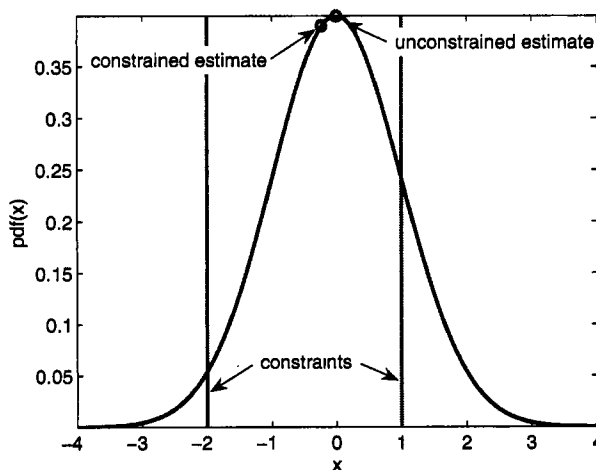


Figure 7.3 The unconstrained estimate at $x = 0$ satisfies the constraints. Nevertheless, the truncation approach to constrained estimation shifts the estimate to the centroid of the truncated pdf, which is at $x \approx -0.23$.